

Conversational Clustering on conflict events

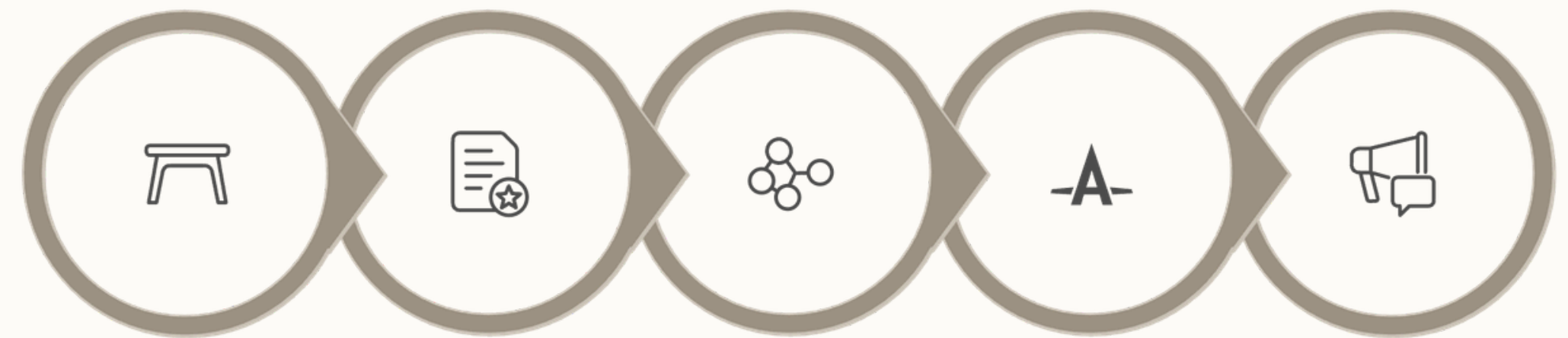
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PROBLEM

- Conflict researchers face thousands of structured event records.
- The goal is to find recurring patterns: where, when, who, what type of violence, and how severe.
- Clustering can reveal groups of similar events, but UCDP GED has no ground-truth cluster labels.
- Raw distance metrics are hard to interpret because the data is categorical, geographic, and imbalanced.

RESEARCH QUESTION:

Can a researcher refine clusters through natural language instead of manually tuning feature weights?



Raw Events

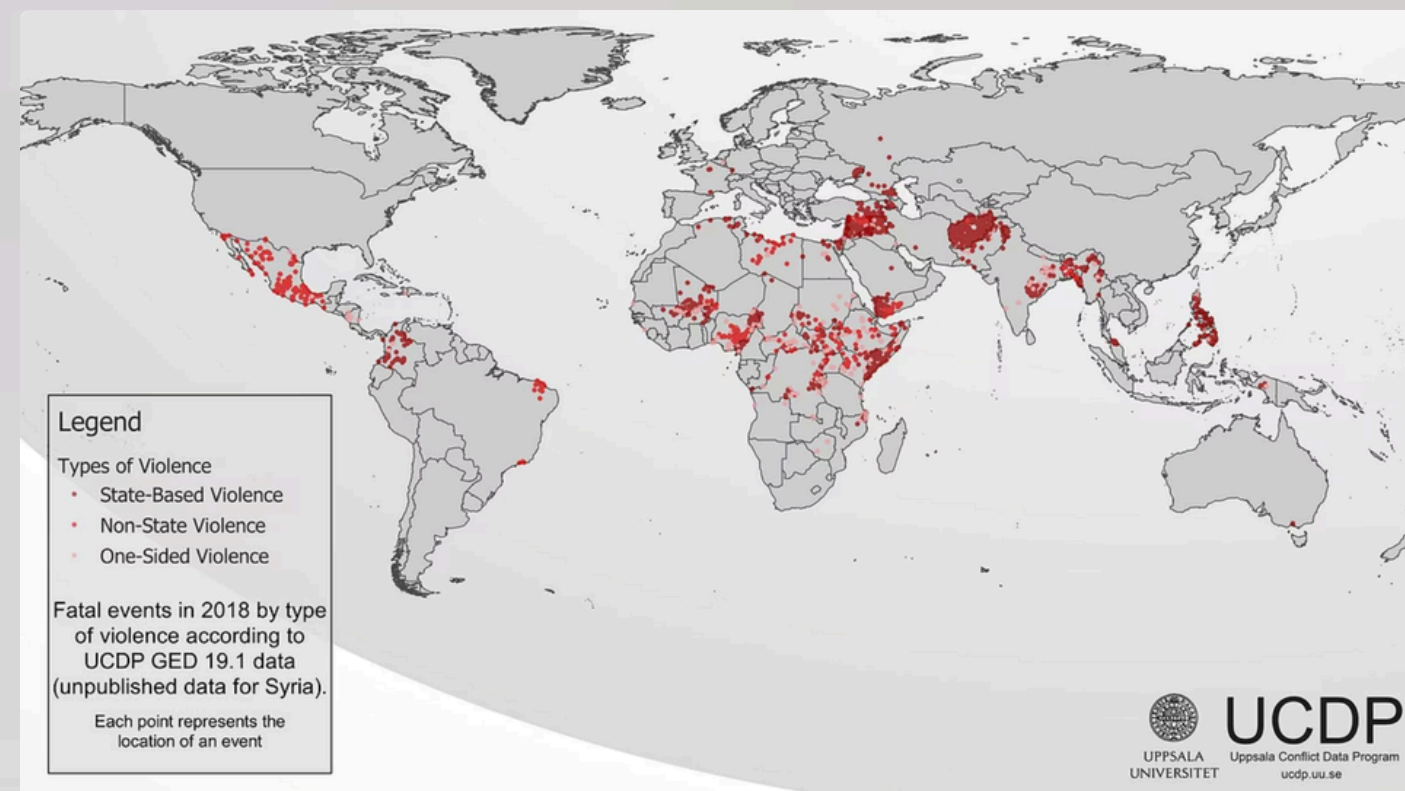
Feature
Space

Clusters

Refinement

Analyst Loop

RESULTS DASHBOARD



What We Actually Tested

H1

Conversational Refinement

Human conversational refinement improves final Silhouette score over one-shot k-means.

H2

Feature Richness

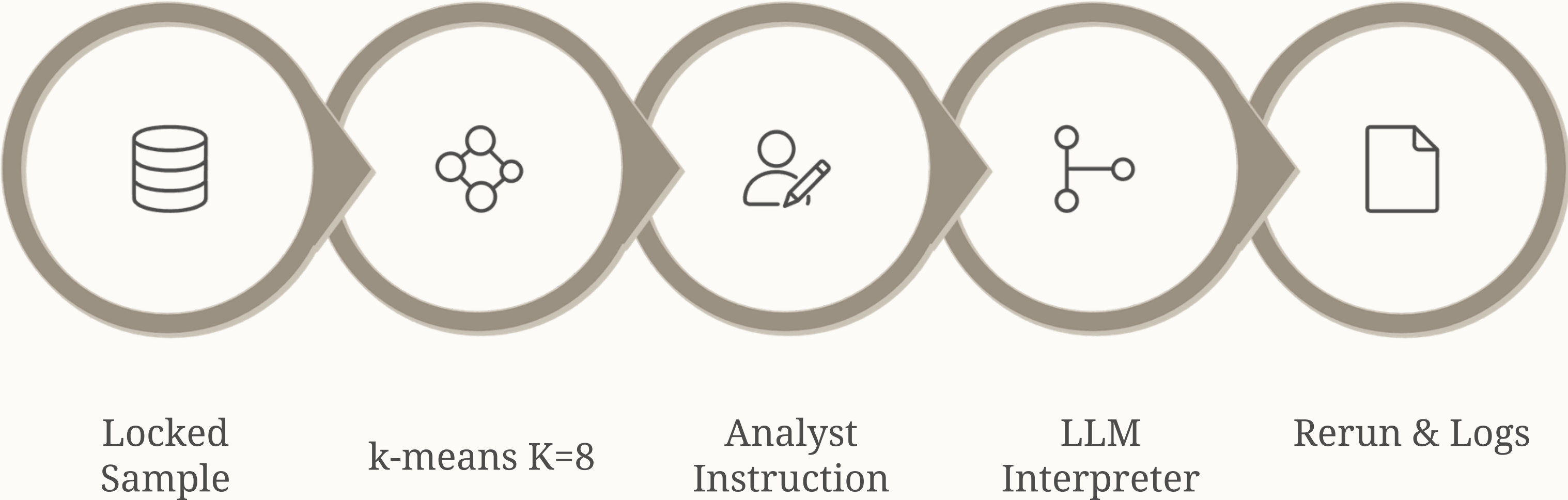
Refinement helps more with richer features (F2) than with location-only features (F1).

H3

LLM Oracle Proxy

An LLM oracle can proxy human judgements of cluster quality.

How the System Works



Three prompt roles drive the system: an **interpreter prompt** converts natural language into JSON feature weights; an **oracle-user prompt** simulates a domain-naïve analyst (Condition C); and an **oracle-eval prompt** judges A/B cluster pairs for H3.

Condition A One-shot baseline	Condition B Human conversational refinement	Condition C Oracle conversational refinement
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Evaluation: Metric Quality + Human Preference

Condition	F1 — Location Only	F2 — Richer Features
A — Baseline	One-shot k-means	One-shot k-means
B — Human	Conversational refinement	Conversational refinement
C — Oracle	Conversational refinement	Conversational refinement

180 Total Runs

3 conditions × 2 feature sets × 30 seeds. K fixed at 8 for comparability.

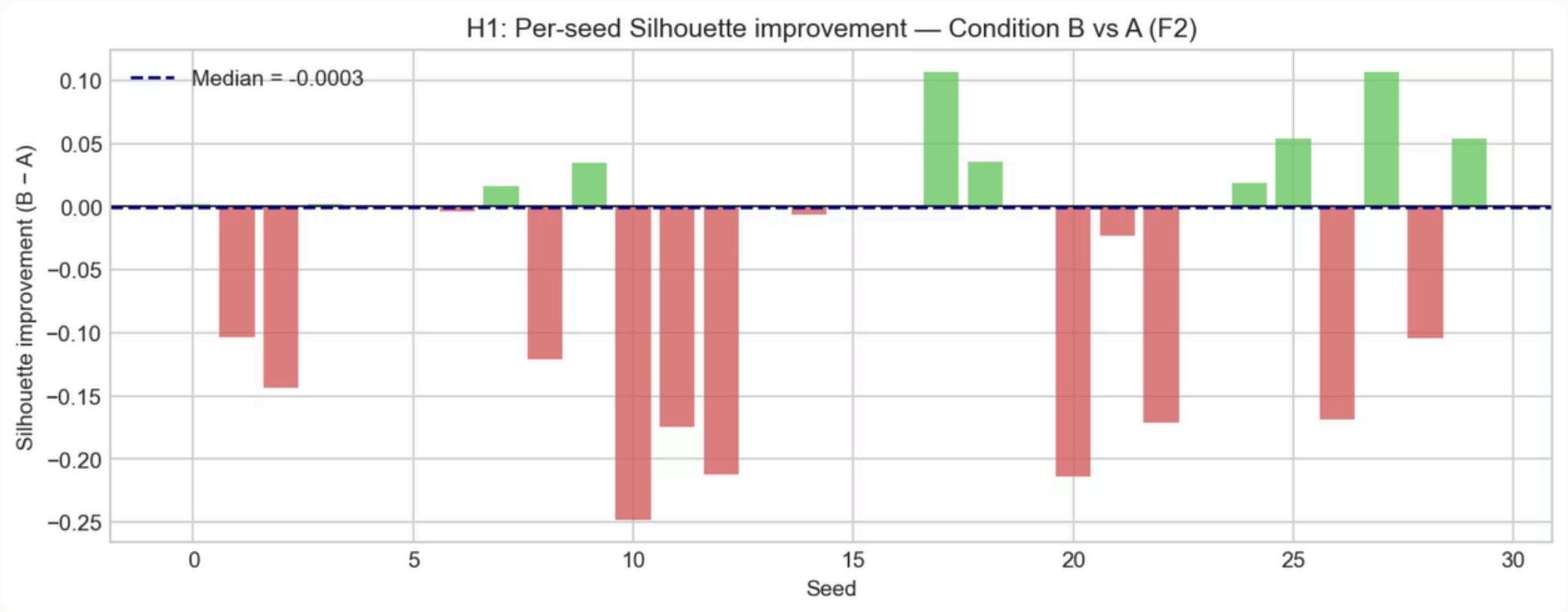
Primary Metric

Final Silhouette score. Davies-Bouldin as secondary.

H3 Design

20 cluster-pair comparisons, 5 human raters, with original and swapped A/B oracle judgements.

H1: Refinement Did Not Clearly Beat Baseline



Condition B minus Condition A under F2, across 30 paired seeds.

-0.0003

Median Improvement

13

Positive Seeds

15 negative, 2 zero



95% CI: [-0.0632, 0.0015] — **H1 not supported.** Conversational refinement changed outcomes, but not consistently upward.

H2: Did Richer Features Help More?

The Hypothesis

- F2 adds actor identity, violence type, and fatality level to F1's location-only features
- More dimensions → more levers for conversational refinement

The Result (Silhouette)

- F1 median improvement (B – A): 0.000
- F2 median improvement (B – A): -0.003
- 95% CI on difference: [-0.034, +0.044] — includes zero

 H2 not supported — no significant difference between F1 and F2 improvement

Why: F2 Was Not Actually Richer Here

The Russia-Ukraine subset has near-constant categorical features.

F2 Extra Feature	Variance
actor_Government of Ukraine	~0
actor_Civilians	~0
type_of_violence (state-based)	~0
type_of_violence (one-sided)	~0

F2 effectively reduces to F1 + fatalities on this dataset. The null result reflects the dataset structure, not a flaw in the method.

A Different Signal: Davies-Bouldin

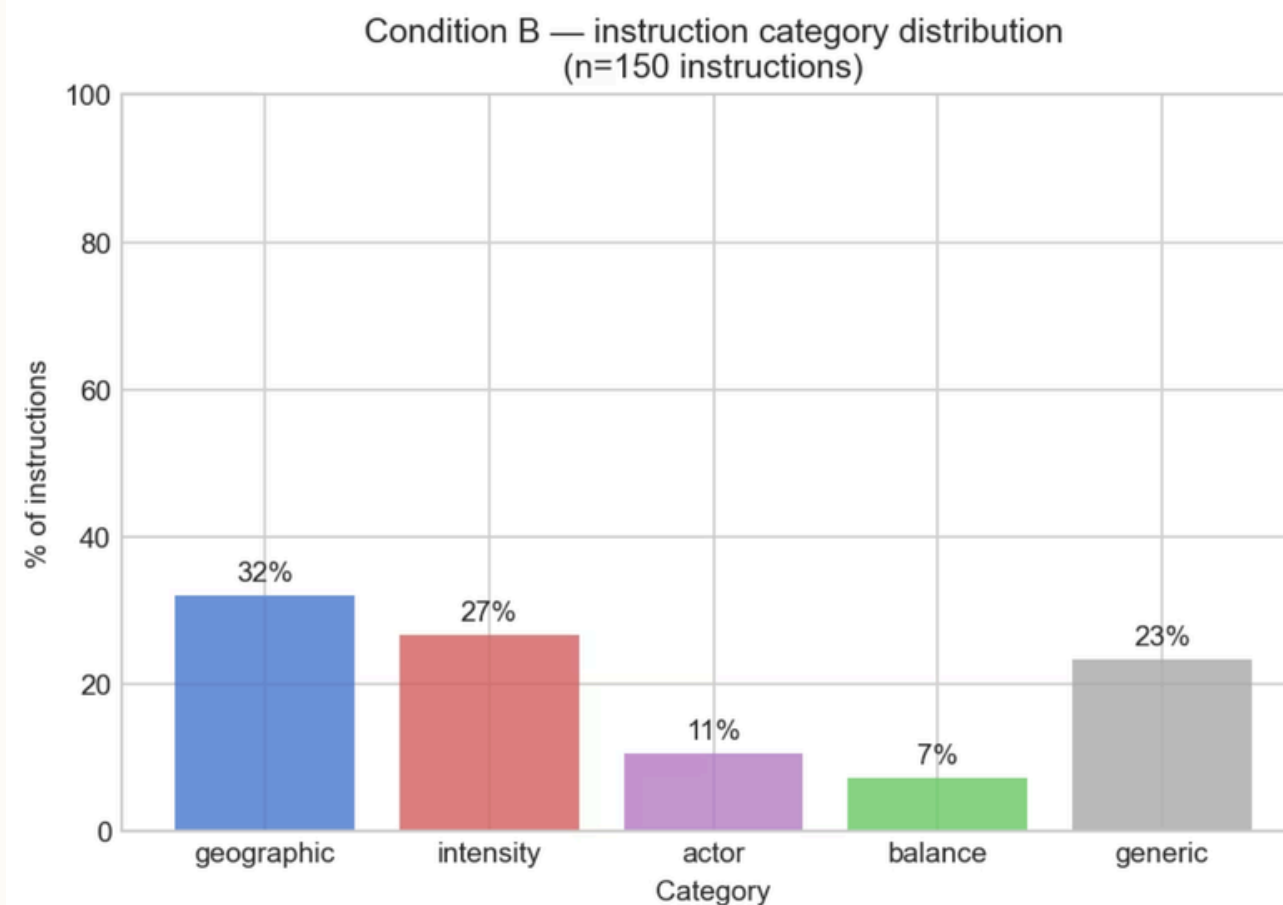
Silhouette — no difference

- F1: 15 / 30 seeds improved
- F2: 13 / 30 seeds improved

Davies-Bouldin — F2 wins

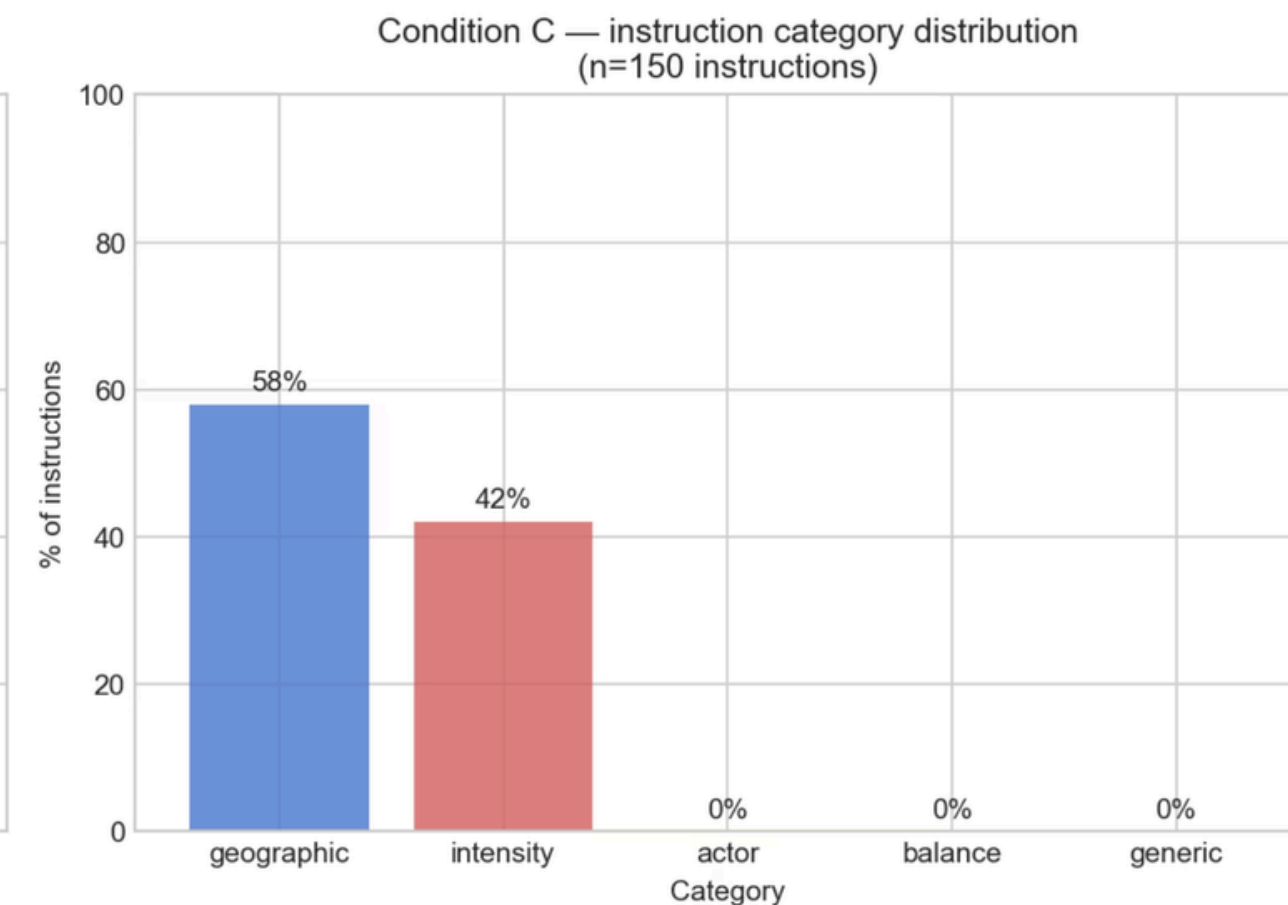
- F1: 15 / 30 seeds improved
- F2: 25 / 30 seeds **improved**

Instruction taxonomy — B vs C / F2 (exploratory, §11.1)



Human instructions — Condition B

Geography · Intensity · Actor · Balance · Generic



Oracle instructions — Condition C

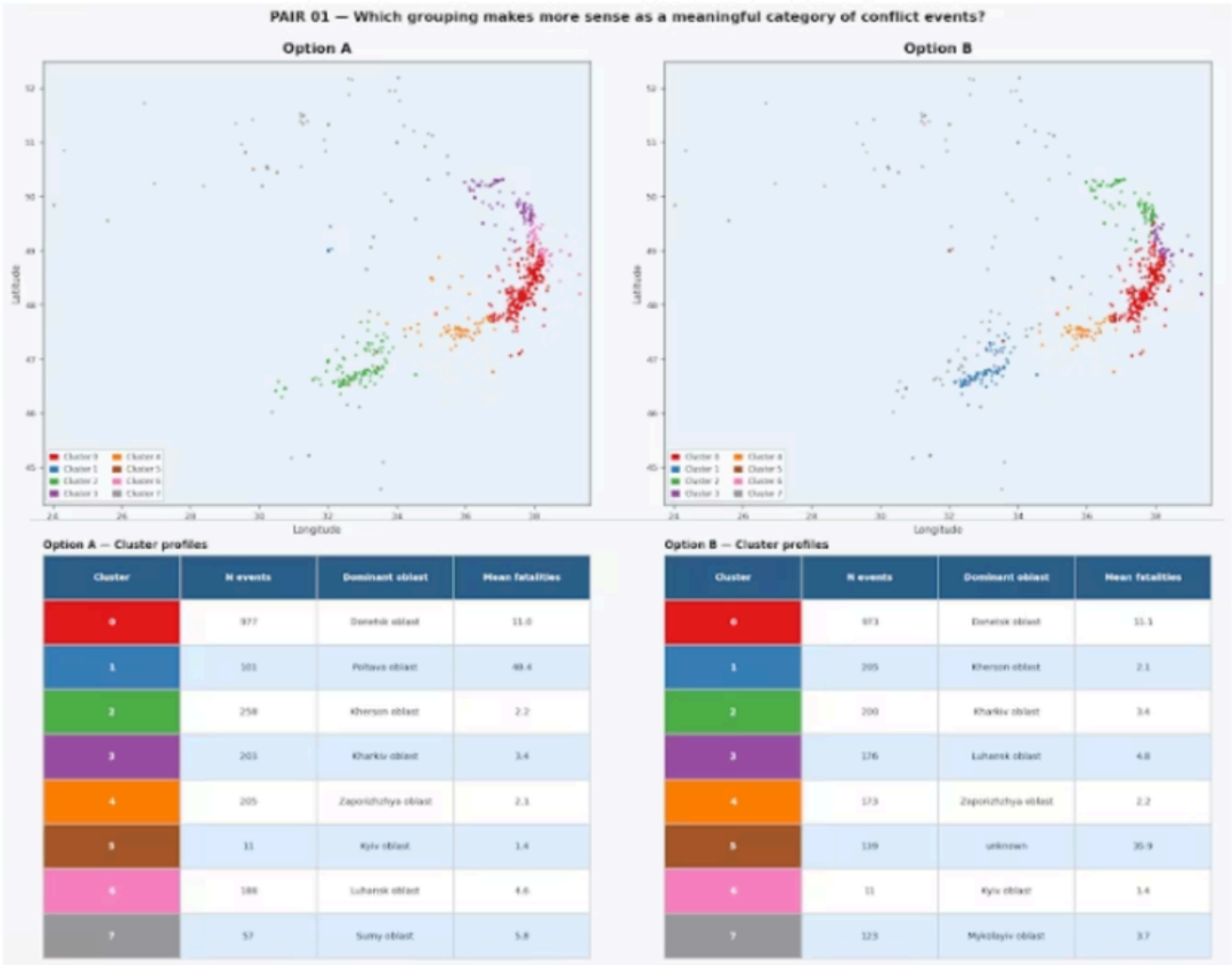
Geography · Intensity only



Humans explored more dimensions in F2 — but the dataset limited the impact. The improvement shows in cluster separation (DB), not cohesion (Silhouette).

H3: Oracle as a Human Proxy

Pair 01 — Which grouping makes more sense as a meaningful category of conflict events?



☐ Option A

☐ Option B

Humans vs LLM oracle on 20 A/B cluster comparisons.

- Same task: choose the more meaningful grouping.
- A/B labels were swapped as a bias check.
- Pairs compared A: baseline, B: human refinement, and C: oracle refinement.

65%

Oracle–Human Agreement
13 of 20 comparisons

What We Learned

What Was Hard

- The data was mostly structured and categorical.
- Little semantic signal for LLM.
- Many variables had low variation, so adding features did not add useful clustering information.

Future work

- Test conversational clustering on datasets where text descriptions play a larger role.
- Start from expert-defined or historically meaningful clusters as reference points.
- Use the conversation loop to help researchers refine, compare, and discover possible cluster structures.

Team Contribution

Giacomo Vettore contributed to the data exploration, experimental execution, report/dashboard preparation, H3 oracle setup and output evaluation, H3 documentation and final synthesis.

Daniele Notarangelo contributed to the initial study design and quality specification, repository structure, core clustering and logging implementation, Condition A and C execution, README/reporting materials, and figure updates.

Jasnoor Singh contributed to the study-plan evolution, related-work synthesis, Condition B execution, H3 pair-generation materials